

Quiz - Lecture 06

Close your laptops and put away all electronics. No electronics or notes are allowed during the duration of this quiz.

Question 1

What is the base case in a recursive function?

- A. The case that runs most often
- B. The case that returns without making a recursive call
- C. The case that uses a loop instead of recursion
- D. The case that produces the final output

Question 2

What happens if a recursive function never reaches a base case?

- A. The function returns `None`
- B. The program prints an error message
- C. The program runs forever or crashes
- D. The function automatically stops

Question 3

What does the function return if it does not have a `return` statement?

- A. `0`
- B. `False`
- C. `None`
- D. An error is raised

Question 4

Which statement about recursion is true?

- A. Every recursive function must use a loop
- B. Recursive functions cannot return values
- C. Recursive functions must call themselves
- D. Recursion is faster than iteration in Python

Question 5

What is the value of y at the end of this code?

```
Python
def function(x):
    if x < 0:
        return -x
    return x

y = -5
y = function(y)
```

- A. 5
- B. -5
- C. None
- D. 0

Question 6

What is the output of this code?

```
Python
def function(a, b):
    return a + b * a + b

print(function(3, 4))
```

- A. function(3, 4)
- B. 19
- C. 7
- D. An error

Question 7

What is the output of this code?

```
Python
x = 4
y = 10

if (x > 5 or y > 5) and x < 10:
    print("A")
else:
    print("B")
```

- A. A
- B. B
- C. Both A and B are printed
- D. Nothing is printed

Question 8

What is the output of this code?

```
Python
x = "12"

if type(x) == str:
    if len(x) == 2:
        x = int(x)
        if type(x) == int:
            print(x + 1)
        else:
            print("Not int")
    else:
        print("Wrong length")
else:
    print("Not string")
```

- A. Not int
- B. 12
- C. 13
- D. Wrong length

Question 9

What best describes what happens when the following code is executed?

```
Python
def recurse(n):
    if n > 0:
        recurse(n + 1)

recurse(1)
```

- A. It prints numbers forever
- B. It stops when n becomes large
- C. It causes a runtime error due to too many recursive calls
- D. It prints None

Question 10

What is the output of this code?

```
Python
x = -1
if x > 0 and x < 10:
    print("A")
elif x > 0 or x == -1:
    print("B")
else:
    print("C")
```

- A. A
- B. B
- C. C
- D. None

Bonus Question 1

What is the output of this code?

```
Python
def function(n):
    x = (n % 2 == 0)
    print(x)

function(5)
```

- A. None
- B. False
- C. 0
- D. True

Bonus Question 2

What is the output of this code?

```
Python
def g(n):
    if n < 0:
        print("neg")
    print("pos")

g(-1), g(0)
```

- A. neg pos neg
- B. neg pos pos
- C. neg pos
- D. -1 0

Bonus Question 3

In your own words, describe what a `RecursionError` is.

Hint: What is the default, maximum recursion depth in Python?

A recursion error occurs when a program reaches the maximum recursion depth (1000 by default). It indicates that a program has exceeded the recursive limit, likely stuck in a recursive loop with no base case.

Bonus Question 4

The following function prints a sequence of numbers using a for loop with `range(start, stop, step)`.

Python

```
def print_range(start, stop, step):  
    for i in range(start, stop, step):  
        print(i)
```

Rewrite this function using recursion instead of a for loop. It should take the same parameters (`start`, `stop`, `step`).

- For when `step` is larger than 0 (this answer would suffice):

```
def print_range(start, stop, step):  
    if start < stop:  
        print(start)  
        print_range(start + step, stop, step)
```

- For when `step` is 0 or less than 0:

```
def print_range(start, stop, step):  
    if step == 0:  
        raise ValueError("range() arg 3 must not be zero")  
    if step > 0 and start < stop:  
        print(start)  
        print_range(start + step, stop, step)  
    if step < 0 and stop < start:  
        print(start)  
        print_range(start + step, stop, step)
```